

Quiz # 3 Solution

(a) Total mass balance $\frac{dm_{sys}}{dt} = \dot{m}_i - \dot{m}_e = (10-15) \frac{kg}{min} = -5 \frac{kg}{min}$
separate and integrate

$$\int_{100 kg}^{m_{sys}} dm_{sys} = -5 \frac{kg}{min} \int_0^t dt = -5 \frac{kg}{min} t$$

$$m_{sys} = 100 kg - 5 \frac{kg}{min} t \quad \text{or} \quad \boxed{m_{sys} = 100 - 5t} \quad \begin{matrix} t \in [0, 20] \text{ min} \\ m_{sys} \in [0, 100] \text{ kg} \end{matrix}$$

(b) $m_{sys} = 100 - 5(10) = \underline{50} \text{ kg}$

(c) Species mass balance on A

$$\frac{dm_{A,sys}}{dt} = \dot{m}_{A,i} - \dot{m}_{A,e}$$

Use $m_{A,sys} = W(t) m_{sys}(t)$

$$\dot{m}_{A,i} = W_i \dot{m}_i$$

$$\dot{m}_{A,e} = W(t) \dot{m}_e$$

$$\frac{d(W m_{sys})}{dt} = W_i \dot{m}_i - W \dot{m}_e$$

$$m_{sys} \frac{dW}{dt} + W \frac{dm_{sys}}{dt} = W_i \dot{m}_i - W \dot{m}_e$$

$$m_{sys} \frac{dW}{dt} = W_i \dot{m}_i - W (\dot{m}_e + \frac{dm_{sys}}{dt})$$

$$(100-5t) \frac{dW}{dt} = (0.1)(10) - W(15-5) = 1 - 10W$$

separate and integrate

$$\int_{0.6}^{W_f} \frac{dW}{1-10W} = \int_0^{10} \frac{dt}{100-5t}$$

$$-\frac{1}{10} \ln \frac{1-10W_f}{1-10(0.6)} = -\frac{1}{5} \ln \frac{100-5(10)}{100}$$

$$\frac{1-10W_f}{-5} = \left(\frac{1}{2}\right)^2$$

$$W_f = \frac{1 + \frac{5}{4}}{10} = \frac{9}{40} = 0.225 \quad \text{or} \quad 22.5 \text{ wt\% A at 10 mins.}$$

General Solution

$$\int_{0.6}^{W(t)} \frac{dW}{1-10W} = \int_0^t \frac{dt}{100-5t}$$

$$\ln \frac{1-10W(t)}{-5} = \ln \left(\frac{100-5t}{100} \right)$$

$$W(t) = \frac{1}{10} + \frac{1}{2} (1-0.05t)^2$$

$t \in [0, 20] \text{ mins.}$

Alternate Solution for part (c)

Write species balance for A and B

$$A: \frac{dm_{A,sys}}{dt} = (0.1)10 - \frac{m_{A,sys}}{m_{sys}} 15$$

$$\text{where } w(t) = \frac{m_{A,sys}}{m_{sys}}$$

$$B: \frac{dm_{B,sys}}{dt} = (0.9)10 - \frac{m_{B,sys}}{m_{sys}} 15$$

$$\text{where } 1-w(t) = \frac{m_{B,sys}}{m_{sys}}$$

$$\frac{d(9m_A - m_B)}{dt} = \frac{(9m_A - m_B)}{m} (15)$$

$$\text{let } u = 9m_A - m_B$$

$$\int_{u_0}^u \frac{du}{u} = -15 \int_0^t \frac{dt}{100-5t}$$

$$\ln \frac{u}{u_0} = \frac{-15}{-5} \ln \left(\frac{100-5t}{100} \right) = \ln(1-0.05t)^3$$

$$u(t) = u_0 (1-0.05t)^3$$

$$9m_A - m_B = [9(60) - 40](1-0.05t)^3 = 500(1-0.05t)^3$$

$$\text{Use } m_A + m_B = 100 - 5t$$

$$10m_A = 100 - 5t + 500(1-0.05t)^3$$

$$m_A = 10 - 0.5t + 50(1-0.05t)^3$$

$$w(t) = \frac{m_A}{m} = \frac{10 - 0.5t + 50(1-0.05t)^3}{100 - 5t}$$

$$w(t) = \frac{1}{10} + \frac{1}{2}(1-0.05t)^2$$

$$w(10) = \frac{1}{10} + \frac{1}{2}\left(\frac{1}{2}\right)^2 = \frac{1}{10} + \frac{1}{8} = \frac{4+5}{40} = \frac{9}{40}$$

The two species balance are not independent. They can be combined to give one separable equation

$$m_{A,sys} \rightarrow m_A \quad m_{B,sys} \rightarrow m_B \\ m_{sys} \rightarrow m$$